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Data management method for semiconductor manufacturing apparatus and control device provided with ring buffer

Abstract

Conventionally, in order to hold data with a high sampling rate in a control device, it has been necessary to increase the capacity of ring buffers. A data management method for a semiconductor manufacturing apparatus includes acquiring, by a control device, data related to operation of the semiconductor manufacturing apparatus, identifying, by the control device, first data and second data of the acquired data, writing, by the control device, the first data to a first ring buffer at first temporal granularity and the second data to a second ring buffer at second temporal granularity coarser than the first temporal granularity, and extracting, by a data management device, the first and second data from the first ring buffer and the second ring buffer respectively and storing the first and second data in a database.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a data management method for a semiconductor manufacturing apparatus and a control device provided with a ring buffer.

BACKGROUND ART

(2) Control devices such as PLC (Programmable Logic Controller) are used to control a semiconductor manufacturing apparatus. Each control device is provided with a ring buffer and stores control data for the semiconductor manufacturing apparatus and measured data from various sensors provided in the semiconductor manufacturing apparatus in the ring buffer in chronological order. The data in the ring buffer is retrieved from the ring buffer by an external computer and

saved in a storage device of the computer. As related technologies, for example, one disclosed in Patent Literature 1 is known.

CITATION LIST

Patent Literature

(3) PTL 1: Japanese Patent Application Laid-Open No. 2004-319857

SUMMARY OF INVENTION

Technical Problem

(4) For accurate management of operating conditions or the like of each section of the semiconductor manufacturing apparatus, it is desirable to increase a sampling rate of the above-described control data and measured data. However, the capacity of the ring buffer needs to be increased for the control device to hold data with a high sampling rate.

Solution to Problem

(5) [Aspect 1]

(6) According to Aspect 1, a data management method for a semiconductor manufacturing apparatus is provided, the method including acquiring, by a control device, data related to operation of the semiconductor manufacturing apparatus, identifying, by the control device, first data and second data of the acquired data, writing, by the control device, the first data to a first ring buffer at first temporal granularity and the second data to a second ring buffer at second temporal granularity coarser than the first temporal granularity, and extracting, by a data management device, the first and second data from the first ring buffer and the second ring buffer respectively and storing the first and second data in a database.

(7) [Aspect 2]

(8) According to Aspect 2, in the method according to Aspect 1, the first data and the second data are written to the first ring buffer and the second ring buffer periodically at the first temporal granularity and the second temporal granularity respectively.

(9) [Aspect 3]

(10) According to Aspect 3, in the method according to Aspect 1, the second data is written to the second ring buffer only when there is a change in the second data.

(11) [Aspect 4]

(12) According to Aspect 4, in any one of the methods according to Aspects 1 to 3, the step of storing the first and second data in the database includes correcting time information of the second data using time information of the first data as a reference and storing the second data with the corrected time information in the database in association with the first data used as the reference for the time information.

(13) [Aspect 5]

(14) According to Aspect 5, in the method according to Aspect 4, one piece of the second data is associated with a plurality of pieces of the first data.

(15) [Aspect 6]

(16) According to Aspect 6, in any one of the methods according to Aspects 1 to 3, the step of storing the first and second data in the database includes correcting time information of the first data using time information of the second data as a reference and storing the first data with the corrected time information in the database in association with the second data used as the reference for the time information.

(17) [Aspect 7]

(18) According to Aspect 7, in the method according to Aspect 6, a selected one of the plurality of pieces of the first data is associated with one piece of the second data.

(19) [Aspect 8]

(20) According to Aspect 8, in any one of the methods according to Aspects 1 to 7, the first data and the second data are measured data measured by a sensor provided in the semiconductor manufacturing apparatus.

(21) [Aspect 9]

(22) According to Aspect 9, in any one of the methods according to Aspects 1 to 7, the first data is measured data measured by a sensor provided in the semiconductor manufacturing apparatus and the second data is data indicating operation contents of the semiconductor manufacturing apparatus.

(23) [Aspect 10]

(24) According to Aspect 10, a data management method for a semiconductor manufacturing apparatus is provided, the method including acquiring, by a first control device, first data related to operation of the semiconductor manufacturing apparatus, acquiring, by a second control device, second data related to operation of the semiconductor manufacturing apparatus, writing, by the first control device, the first data to a first ring buffer at first temporal granularity, writing, by the second control device, the second data in a second ring buffer at second temporal granularity coarser than the first temporal granularity, extracting, by a data management device, the first and second data from the first ring buffer and the second ring buffer respectively, correcting, by the data management device, time information of the second data using time information of the first data as a reference and storing, by the data management device, the second data with the corrected time information in a database in association with the first data used as a reference for the time information.

(25) [Aspect 11]

(26) According to Aspect 11, a control device connected to a semiconductor manufacturing apparatus is provided, the control device including a first ring buffer and a second ring buffer, in which the control device is configured to acquire data related to operation of the semiconductor manufacturing apparatus, identify first data and second data in the acquired data, write the first data to a first ring buffer at first temporal granularity and write the second data to a second ring buffer at second temporal granularity coarser than the first temporal granularity.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a configuration diagram of a system that implements a method according to an embodiment of the present invention:

(2) FIG. 2 is a flowchart illustrating an overall outline of a process to implement the method according to the embodiment of the present invention:

(3) FIG. 3 is a diagram illustrating details of a process in step S204 according to the embodiment of the present invention and illustrating an example of high speed sampling data written to a first ring buffer:

(4) FIG. 4 is a diagram illustrating details of the process in step S204 according to the embodiment of the present invention and illustrating an example of low speed sampling data written to a second ring buffer;

(5) FIG. 5 illustrates a conventional example contrasted with FIG. 3 and FIG. 4;

(6) FIG. 6 is a diagram illustrating details of the process in step S204 according to another embodiment of the present invention and illustrating an example of measured data written to the second ring buffer;

(7) FIG. 7 is a diagram illustrating details of the process in step S204 according to another embodiment of the present invention and illustrating an example of event data written to a third ring buffer:

(8) FIG. 8 illustrates a conventional example contrasted with FIG. 6 and FIG. 7;

(9) FIG. 9 is a diagram illustrating an example of a process of writing event data to the third ring buffer:

- (10) FIG. 10 is a diagram illustrating an example of a process of writing control bit data to the third ring buffer;
- (11) FIG. 11 is a diagram illustrating details of a process in step S205 according to the embodiment of the present invention and illustrating an example of a process of temporally matching high speed sampling data in the first ring buffer and low speed sampling data in the second ring buffer;
- (12) FIG. 12 is a diagram illustrating details of the process in step S205 according to the embodiment of the present invention and illustrating an example of a process of temporally matching measured data in the second ring buffer and event data in the third ring buffer;
- (13) FIG. 13 is a diagram illustrating details of the process in step S205 according to the embodiment of the present invention and illustrating another example of the process of temporally matching measured data in the second ring buffer and event data in the third ring buffer;
- (14) FIG. 14 is a diagram illustrating another configuration example of a system that implements the method according to the embodiment of the present invention; and
- (15) FIG. 15 is a diagram illustrating details of the process in step S205 applicable to the system in FIG. 14.

DESCRIPTION OF EMBODIMENTS

- (16) Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings.
- (17) FIG. 1 is a configuration diagram of a system 10 that implements a method according to an embodiment of the present invention. The system 10 is provided with a semiconductor manufacturing apparatus 100, a control device 200 and a data management device 300. The semiconductor manufacturing apparatus 100 is an apparatus to manufacture a semiconductor product or an intermediate product of the semiconductor product. For example, the semiconductor manufacturing apparatus 100 includes any device involved in manufacturing a semiconductor product such as an electroplating apparatus for plating a substrate (semiconductor substrate or glass substrate or the like), a polishing apparatus (e.g., CMP (Chemical Mechanical Polishing) apparatus) for polishing a substrate or a thin film formed on a substrate, a film forming apparatus (e.g., CVD (Chemical Vapor Deposition) apparatus or a vapor deposition apparatus) for forming a thin film on a substrate, an exposure apparatus for transferring a fine pattern to a thin film on a substrate, an etching apparatus for micromachining a substrate or a thin film on a substrate by etching, an ion implantation apparatus for implanting ions into a substrate or a thin film on a substrate, a dicing apparatus for cutting a substrate into chips, various inspection apparatuses (or measurement apparatuses) for inspecting a final semiconductor product or intermediate product (including substrate).
- (18) The semiconductor manufacturing apparatus 100 is provided with a plurality of drive sections 120, operations of which are controlled by the control device 200 during operation of the semiconductor manufacturing apparatus 100. For example, if the semiconductor manufacturing apparatus 100 is an electroplating apparatus, the drive section 120 includes a transfer device or a transfer robot 120 for transferring a substrate, a chemical supply device 120 for supplying predetermined chemical to each bath such as a plating bath, a plurality of valves 120 that open/close a supply pipe of each chemical, a power supply device (rectifier or the like) 120 that supplies a current for electrolytic reaction to a substrate immersed in a plating solution in a plating bath, a temperature control device 120 that controls temperature of the plating solution in the plating bath and other devices/mechanisms (e.g., alarm device) 120. On the other hand, for example, if the semiconductor manufacturing apparatus 100 is a CMP device, the drive section 120 includes a motor 120 for rotating a substrate and a polishing pad, an opening/closing valve 120 for a pipe that supplies a polishing liquid to a polishing pad, a substrate transfer robot 120 and other devices/mechanisms 120. These are nothing but specific examples of some of many drive sections 120 provided in the semiconductor manufacturing apparatus 100. The semiconductor manufacturing apparatus 100 may be provided with any number of drive sections 120 of any

appropriate type.

(19) The semiconductor manufacturing apparatus **100** is further provided with a plurality of sensors **140**. The plurality of sensors **140** are installed at predetermined locations of the semiconductor manufacturing apparatus **100**. The sensor **140** may be, for example, a temperature sensor, a flow rate sensor, a pressure sensor, a concentration sensor, a torque meter and a rotation speed meter for measuring the torque and the number of revolutions of a motor, an ammeter, a voltmeter or the like. Each sensor **140** is installed at each drive section **120** of the semiconductor manufacturing apparatus **100**, and measures each predetermined physical quantity and outputs the measured value.

(20) The control device **200** is a device that controls and manages operation of the semiconductor manufacturing apparatus **100**. For example, a PLC (Programmable Logic Controller) can be preferably used as the control device **200**, but the control device **200** can be another type of computer. In the system **10** in FIG. **1**, the control device **200** is connected to each drive section **120** and each sensor **140** of the semiconductor manufacturing apparatus **100** via communication wiring **400**. The control device **200** may be assembled in the semiconductor manufacturing apparatus **100** as part of the configuration of the semiconductor manufacturing apparatus **100**.

(21) Note that although only one control device **200** is shown in FIG. **1**, the system **10** may be provided with a plurality of control devices **200**. An embodiment in which the system **10** is provided with the plurality of control devices **200** will be described later with reference to FIG. **14** or the like.

(22) The control device **200** is provided with a processor **202**, a memory for program storage **203**, a plurality of ring buffers **204-1** to **204-4**, an input/output interface **205**, a communication interface **206** and a timer **207**. A program for implementing the method (or part of the method) according to the embodiment of the present invention is stored in the memory **203**. The processor **202** reads and executes the program from the memory **203**. Thus, the control device **200** can implement the method (or part of the method) according to the embodiment of the present invention.

(23) The processor **202** of the control device **200** outputs an operation instruction to each drive section **120** of the semiconductor manufacturing apparatus **100** via the input/output interface **205**. According to this operation instruction, each drive section **120** of the semiconductor manufacturing apparatus **100** performs a predetermined operation. The processor **202** of the control device **200** acquires a measured value measured by each sensor **140** from each sensor **140** of the semiconductor manufacturing apparatus **100** via the input/output interface **205** and writes data with the acquired measured value to each of the ring buffers **204-1** to **204-4**.

(24) Furthermore, in the system **10** in FIG. **1**, the control device **200** is connected to the data management device **300** via a network **500** such as a LAN (local area network) or the Internet. The control device **200** can communicate with the data management device **300** via the communication interface **206**.

(25) In the system **10** of the present embodiment, the data management device **300** is a device that has a role of managing data related to operation of the semiconductor manufacturing apparatus **100**. The data management device **300** may be, for example, a computer provided with a data server function. The data management device **300** is provided with a processor **302**, a memory for program storage **303** and a database **304**. A program for implementing the method (or part of the method) according to the embodiment of the present invention is stored in the memory **303**. The processor **302** reads and executes a program from the memory **303**. Thus, the data management device **300** can implement the method (or part of the method) according to the embodiment of the present invention.

(26) FIG. **2** is a flowchart illustrating an overall outline of a process to implement the method according to the embodiment of the present invention. The processing of the present embodiment includes operations of the semiconductor manufacturing apparatus **100**, the control device **200** and the data management device **300**.

(27) First, in step **S201**, the control device **200** outputs a predetermined operation instruction to

each drive section **120** of the semiconductor manufacturing apparatus **100**. In step **S202**, each drive section **120** of the semiconductor manufacturing apparatus **100** performs a predetermined operation according to an operation instruction instructed by the control device **200**. In step **S203**, each sensor **140** of the semiconductor manufacturing apparatus **100** measures data and transmits the measured value to the control device **200**.

(28) In step **S204**, the control device **200** receives the measured values transmitted from each sensor **140** of the semiconductor manufacturing apparatus **100** and temporarily stores the measured values in the ring buffers **204-1** to **204-4**. After step **S204**, steps **S201** to **S204** are repeated again at predetermined time intervals.

(29) In step **S205**, the data management device **300** extracts data from the ring buffers **204-1** to **204-4** of the control device **200** and stores the data in the database **304**. In this way, data related to operation of the semiconductor manufacturing apparatus **100** is accumulated in the database **304** of the data management device **300**. The data stored in the database **304** may then be used in the data management device **300** for purposes such as, displaying the data on a monitor screen for an administrator of the system **10** to check the operational status (e.g., occurrence of trouble) of the semiconductor manufacturing apparatus **100** or further analyzing the operational status of the semiconductor manufacturing apparatus **100** (e.g., statistical process).

(30) FIG. **3** and FIG. **4** are diagrams illustrating details of the process in above-described step **S204** according to the embodiment of the present invention. In the present embodiment, the control device **200** holds data that needs to be recorded with a high sampling rate (hereinafter referred to as “high speed sampling data”) of the measured values obtained from each sensor **140** in the first ring buffer **204-1**, on one hand, and holds data for which recording at a low sampling rate is sufficient (hereinafter referred to as “low speed sampling data”) in the second ring buffer **204-2**, on the other hand.

(31) Whether the measured value of each sensor **140** is high speed sampling data or low speed sampling data may be, for example, determined in advance by the administrator of the system **10** and may be set in the control device **200**. For example, the high speed sampling data may be data that directly indicates the operational status of the drive section **120** of the semiconductor manufacturing apparatus **100** (e.g., the torque and the number of revolutions of a motor, position information or speed information on a transfer device, a current value and a voltage value of each circuit or the like). The low speed sampling data may be data of an environment that changes by being indirectly affected by operation of the drive section **120** of the semiconductor manufacturing apparatus **100** (e.g., temperature, pressure, concentration, flow rate or the like).

(32) In step **S204**, the processor **202** of the control device **200** classifies each sensor **140** of the semiconductor manufacturing apparatus **100** into a sensor **140** corresponding to the high speed sampling data and a sensor **140** corresponding to the low speed sampling data. This classification can be performed based on, for example, information set by the administrator of the system **10**. The processor **202** acquires measured data from the sensor **140** corresponding to the high speed sampling data and writes the measured data to the first ring buffer **204-1** at first temporal granularity and acquires measured data from the sensor **140** corresponding to the low speed sampling data and writes the measured data in the second ring buffer **204-2** at second temporal granularity. Furthermore, when writing the measured data to each ring buffer, the processor **202** acquires date and time information (the time of writing) from the timer **207** and stores the date and time information in the ring buffer in association with the measured data.

(33) The first temporal granularity may be, for example, a 10-millisecond time interval. On the other hand, the second temporal granularity may be temporal granularity coarser than the first temporal granularity, for example, a 100-millisecond time interval. That is, the processor **202** in this example writes the high speed sampling data to the first ring buffer **204-1** once every 10 milliseconds and writes the low speed sampling data to the second ring buffer **204-2** once every 100 milliseconds. Note that numerical values of these time intervals (10 milliseconds, 100

milliseconds) are nothing but examples, and it goes without saying that the numerical values of these time intervals can be changed as appropriate.

(34) FIG. 3 illustrates an example of high speed sampling data written to the first ring buffer **204-1**. In FIG. 3, measured data **D1** is acquired from a predetermined sensor (sensor corresponding to the high speed sampling data) of the plurality of sensors **140** of the semiconductor manufacturing apparatus **100** periodically at times **t010**, **t020**, **t030**, . . . and written to the first ring buffer **204-1**. Times **t010**, **t020**, **t030**, . . . correspond to a 10-millisecond interval. Each measured data **D1** is stored in association with date and time information indicating respective writing times.

(35) On the other hand, FIG. 4 illustrates an example of low speed sampling data written to the second ring buffer **204-2**. In FIG. 4, measured data **D2** and **D3** are acquired from a predetermined sensor (sensor corresponding to the low speed sampling data) of the plurality of sensors **140** of the semiconductor manufacturing apparatus **100** periodically at times **t000**, **t100**, **t200**, . . . and written to the second ring buffer **204-2** together with date and time information. Times **t000**, **t000**, **t200**, . . . correspond to a 100-millisecond interval. Note that when there are a plurality of pieces of data (**D2**, **D3**) as in this example, the plurality of pieces of data are written to the ring buffer together as one set of data.

(36) Here, the present embodiment is contrasted with a conventional example. FIG. 5 illustrates a conventional example in which all the measured data acquired from the plurality of sensors **140** of the semiconductor manufacturing apparatus **100** are written to one ring buffer. As shown in FIG. 5, the measured data **D1**, **D2** and **D3** are acquired from the sensor **140** corresponding to the respective data at each of the times **t010**, **t020**, **t030**, . . . , **t100**, **t110**, . . . with a 10-millisecond interval and written to one and the same ring buffer. When only one ring buffer is used in this way, all the data (**D1**, **D2**, **D3**) are acquired from each sensor and written collectively to the ring buffer (similar to data **D2**, **D3** in the example in FIG. 4).

(37) In the example in FIG. 5, if the measured data **D1** is data that needs to be recorded with a high sampling rate (once every 10 milliseconds) and the measured data **D2** and **D3** are data for which recording at a low sampling rate (e.g., every 100 milliseconds) is sufficient, the measured data **D2** and **D3** at times **t010**, **t020**, **t030**, **t110** or the like are data that need not be recorded originally. However, because of the need to record the measured data **D1** once every 10 milliseconds, the measured data **D2** and **D3** also need to be written to the ring buffer at the same cycle as the measured data **D1**. Thus, in the conventional example, the ring buffer is required to have a buffer size larger than the actual buffer size required so as to be able to hold data that need not be recorded originally.

(38) On the other hand, in the present embodiment, only high speed sampling data (e.g., **D1**) having the first temporal granularity (e.g., 10-millisecond time interval) is written to the first ring buffer **204-1**, whereas low speed sampling data (e.g., **D2**, **D3**) having second temporal granularity (e.g., 100-millisecond time interval) coarser than the first temporal granularity are written to the second ring buffer **204-2**. For this reason, it is sufficient for the first ring buffer **204-1** to have a buffer size necessary to hold only the high speed sampling data (e.g., **D1**) with the first temporal granularity. On the other hand, since the second ring buffer **204-2** holds data with coarse temporal granularity, the buffer size can be small. Therefore, according to the present embodiment, it is possible to reduce the overall buffer size of the ring buffer (total buffer size of the first ring buffer **204-1** and the second ring buffer **204-2**).

(39) FIG. 6 and FIG. 7 are diagrams illustrating details of the process in above-described step **S204** according to another embodiment of the present invention. In the present embodiment, the control device **200** holds measured data obtained from the sensor **140** of the semiconductor manufacturing apparatus **100** in the second ring buffer **204-2** and “event data” in the third ring buffer **204-3**. Note that the measured data from the sensor **140** may be held in the first ring buffer **204-1** instead of the second ring buffer **204-2**.

(40) Here, the event data is data included in an operation instruction sent from the control device

200 to the semiconductor manufacturing apparatus **100** in aforementioned step **S201** in FIG. 2, and is information that specifies the process to be performed by the semiconductor manufacturing apparatus **100** and an object to be processed. For example, when the semiconductor manufacturing apparatus **100** is an electroplating device, the event data includes a step number that identifies a step in a recipe that specifies contents of a series of plating processes made up of a plurality of steps, a work number that identifies a substrate (or a cassette that stores a plurality of substrates) subject to electroplating.

(41) In step **S204** of the flowchart in FIG. 2, the processor **202** of the control device **200** acquires measured data from the sensor **140** of the semiconductor manufacturing apparatus **100** at the first temporal granularity, writes the measured data to the second ring buffer **204-2** (or the first ring buffer **204-1**) on one hand, and writes event data to the third ring buffer **204-3** at the second temporal granularity. Note that the event data can be extracted from the operation instruction generated by the processor **202** in step **S201**. When writing the measured data and the event data to the ring buffer, the processor **202** stores each data in the ring buffer in association with date and time information (the time of writing) from the timer **207**.

(42) The first temporal granularity in the present embodiment may be, for example, a 100-millisecond time interval (or any time interval, for example, several hundreds of milliseconds or less). On the other hand, the second temporal granularity may be, for example, granularity indicated by timing at which the event data changes. Since the event data normally changes (e.g., change in the step number of a recipe or the work number) at a frequency of once every several seconds to every several hundreds of seconds, the second temporal granularity is temporal granularity coarser than the first temporal granularity. That is, in this example, the processor **202** writes measured data from the sensor **140** to the second ring buffer **204-2** once every 100 milliseconds and writes the event data to the third ring buffer **204-3** at timing at which the event data changes.

(43) FIG. 6 illustrates an example of the measured data written to the second ring buffer **204-2**. In FIG. 6, measured data **D2** and **D3** are acquired from a predetermined sensor among the plurality of sensors **140** of the semiconductor manufacturing apparatus **100** periodically at times **t000**, **t100**, **t200**, . . . and written to the second ring buffer **204-2**. Times **t000**, **t100**, **t200**, . . . correspond to a 100-millisecond interval.

(44) On the other hand, FIG. 7 illustrates an example of the event data written to the third ring buffer **204-3**. In FIG. 7, the event data **D1** changes at times **t000**, **t100**, **t300**, **t411**, **t500**, . . . and the event data **D1** is written to the third ring buffer **204-3** aperiodically at these timings.

(45) FIG. 8 is a diagram illustrating a conventional example contrasted with the embodiment in FIGS. 6 and 7, and illustrates an example in which all the measured data and event data are written to one ring buffer. As shown in FIG. 8, the event data **D1** and the measured data **D2** and **D3** are written to the one and the same ring buffer at each of the times **t000**, **t100**, **t200**, **t300**, **t400**, **t411**, **t500**, **t600**,

(46) Here, since the event data **D1** has not changed at times **t200**, **t400**, and **t600**, the event data **D1** at these times are data that need not be recorded originally. Since the time **t411** is not timing of once every 100 milliseconds at which the measured data **D2** and **D3** should be recorded, the measured data **D2** and **D3** at this time **t411** are also data that need not be recorded originally. Therefore, as in the case of FIG. 5, in this conventional example as well, the ring buffers actually need to have buffer sizes larger than the actual buffer size required.

(47) In contrast, according to the embodiment in FIGS. 6 and 7, the measured data **D2** and **D3** are written to the second ring buffer **204-2** at the first temporal granularity (e.g., 100-millisecond time interval) and the event data **D1** is written to the third ring buffer **204-3** at the second temporal granularity (e.g., timing at which the event data **D1** changes) coarser than the first temporal granularity. Therefore, as in the case of the aforementioned embodiment in FIGS. 3 and 4, the overall buffer size of the ring buffer (total buffer size of the second ring buffer **204-2** and the third

ring buffer **204-3**) can be reduced.

(48) FIG. **9** is a diagram illustrating an example of a process of writing only changed event data to the third ring buffer **204-3**. As shown in the upper part of FIG. **9**, the processor **202** of the control device **200** generates a plurality of pieces of event data **D11**, **D12**, **D13**, **D14**, **D15**, **D16** periodically at times **t999**, **t000**, **t001**, . . . , **t006**, . . . , which are predetermined timings and outputs an operation instruction including these event data to the drive section **120** of the semiconductor manufacturing apparatus **100**. In the example in FIG. **9**, although times **t999**, **t000**, **t001**, . . . correspond to a 1-millisecond interval, these times may correspond to any other time intervals.

(49) Here, at time **t000**, event data **D12** changes in value from 5 to 3, and at time **t002**, the event data **D12** changes in value from 3 to 5. Furthermore, the event data **D13** changes in value from 356 to 352, and further at time **t006**, the event data **D14** changes in value from 0 to 1. No event data changes at other times. The processor **202** of the control device **200** writes event data **D11**, **D12**, **D13**, **D14**, **D15**, and **D16** together with date and time information to the third ring buffer **204-3** at these times **t000**, **t002**, **t006** (, . . .)(lower part of FIG. **9**). Note that although the time interval at which the event data changes is 2 milliseconds or 4 milliseconds in FIG. **9**, this is just for convenience of explanation, and the event data actually changes at the second temporal granularity coarser than the first temporal granularity as described above. Although only one event data **D1** is shown in aforementioned FIG. **7**, it goes without saying that there can be a plurality of pieces of event data as shown in FIG. **9**.

(50) FIG. **10** is a diagram illustrating an example of a process of writing “control bit data” to a fourth ring buffer **204-4** using a technique similar to the technique for event data. The control bit data is data included in an operation instruction sent from the control device **200** to the semiconductor manufacturing apparatus **100** in aforementioned step **S201** in FIG. **2** as in the case of the event data and is information that specifies an operating condition (e.g., on or off) of each drive section **120** or each sensor **140** of the semiconductor manufacturing apparatus **100** with bit “0” or “1.” The control bit data is expressed as a bit string with a plurality of consecutive 0s and 1s, and, for example, the first bit corresponds to a sensor A, the second bit corresponds to a sensor B, the third bit corresponds to a valve C, the fourth bit corresponds to a valve D and the fifth bit corresponds to an alarming device E. For example, control bit data “10011 . . . ” means operating conditions such as sensor A: measurement done, sensor B: measurement stopped, valve C: closed, valve D: open, alarming device E: alarm on.

(51) The processor **202** of the control device **200** writes the control bit data to the fourth ring buffer **204-4** at timing when each bit changes. As shown in an upper part of FIG. **10**, the processor **202** generates the control bit data periodically at, for example, times **t999**, **t000**, **t001**, . . . , **t006**, . . . , which is a 1-millisecond interval, and outputs the control bit data as an operation instruction (or part of an operation instruction) to the semiconductor manufacturing apparatus **100**. Here, the control bit data has changed on the ninth bit at time **t000**, on the second bit at time **t002** and on the ninth and tenth bits at time **t006**. No bit has changed at other times. The processor **202** writes the control bit data together with date and time information to the fourth ring buffer **204-4** at times **t000**, **t002**, **t006** (, . . .) (lower part of FIG. **10**).

(52) FIGS. **11** to **13** are diagrams illustrating details of a process in step **S205** of the flowchart in FIG. **2** according to the embodiment of the present invention. As described above, in step **S205**, the data management device **300** extracts data from the ring buffers **204-1** to **204-4** of the control device **200** and stores the data in the database **304**. Here, as has been described so far, data differing in temporal granularity are held in each of the ring buffers **204-1** to **204-4**. The data management device **300** stores data of the respective ring buffers **204-1** to **204-4** as they are in the database **304** and performs a process of temporally matching the data in the database **304** at an appropriate point in time (e.g., at the time of data monitoring or analysis) after that. Alternatively, prior to storing the data extracted from the respective ring buffers **204-1** to **204-4** in the database **304**, the data management device **300** performs a process of temporally matching the data.

(53) FIG. 11 illustrates an example of the process of temporally matching the high speed sampling data in the first ring buffer **204-1** with the low speed sampling data in the second ring buffer **204-2**. With reference to the upper part of FIG. 11, the first ring buffer **204-1** holds high speed sampling data **D1** at times **t000**, **t010**, **t020**, . . . , **t090**, **t100**, **t110**, . . . (10-millisecond interval) and the second ring buffer **204-2** holds low speed sampling data **D10** at times **t000A**, **t100A**, **t200A**, . . . (100-millisecond interval). The processor **302** of the data management device **300** corrects date and time information of the low speed sampling data using the date and time information of the high speed sampling data (that is, data with fine temporal granularity) as a reference. The database **304** stores the low speed sampling data with the corrected date and time information in association with the high speed sampling data used as a reference for correction.

(54) More specifically, in the example in FIG. 11 (however, it is assumed that there is a relationship: $t000A \leq t000$ and $t090 < t100A \leq t100$), the low speed sampling data **D10** at time **t000A** is duplicated into 10 pieces of data having the same date and time information as each high speed sampling data **D1** at reference times **t000**, **t010**, **t020**, . . . , and **t090** (that is, date and time information of the data **D10** is corrected in 10 ways), these 10 pieces of duplicated low speed sampling data **D10** are stored in the database **304** in association with the high speed sampling data **D1** at corresponding reference times. Similarly, the low speed sampling data **D10** at time **t100A** is duplicated into 10 pieces of data having the same date and time information as each high speed sampling data **D1** at reference times **t100**, **t110**, . . . , and **t190** (that is, date and time information is corrected in 10 ways), data **D1** and **D10** at the same date and time are associated with each other and stored in the database **304**. Thus, the database **304** can hold a plurality of pieces of measured data with uniform temporal granularity. This makes it easier to monitor and analyze the measured data.

(55) Note that the date and time information of the low speed sampling data is corrected using the date and time information of the high speed sampling data (that is, data with fine temporal granularity) as a reference in FIG. 11. On the contrary, the date and time information of the high speed sampling data may be corrected using the date and time information of the low speed sampling data (that is, data with coarse temporal granularity) as a reference (hereinafter, see an example in FIG. 13).

(56) FIG. 12 is a diagram illustrating an example of a process of temporally matching measured data in the second ring buffer **204-2** (e.g., low speed sampling data) with event data in the third ring buffer **204-3**. In the upper part of FIG. 12, the second ring buffer **204-2** holds measured data **D2** at times **t100**, **t200**, **t300**, . . . (100-millisecond interval) and the third ring buffer **204-3** holds event data **D11** to **D14** at times **t923**, **t107**, **t801**, . . . (aperiodic). However, a relationship among the respective times is assumed to be $t923 < t100 < t107 < t200 < t300 < t801$. As described above, the temporal granularity of the event data is normally coarser than the temporal granularity of the measured data. The processor **302** of the data management device **300** corrects the date and time information of the event data using the date and time information of the measured data (that is, data with fine temporal granularity) as a reference. The event data with corrected date and time information is stored in the database **304** in association with the measured data used as a reference for correction.

(57) More specifically, in the example in FIG. 12, the date and time information of the event data **D11** to **D14** at time **t923** is corrected to reference time **t100**, associated with the measured data **D2** at the reference time **100**, and stored in the database **304**. On the other hand, the date and time information of the event data **D11** to **D14** at time **t107** (since $t300 < t801$) is corrected in two ways: reference time **t200** and reference time **300** (that is, duplicated into two pieces of data differing in date and time information), associated with the measured data **D2** at time **t200** and time **t300** respectively, and stored in the database **304**. Though not shown, the same applies to event data **D11** to **D14** at time **t801**. In this way, the database **304** can hold measured data and event data with uniform temporal granularity. This makes it easier to monitor and analyze measured data and event

data.

(58) FIG. 13 is a diagram illustrating another example of the process of temporally matching the measured data in the second ring buffer **204-2** (e.g., low speed sampling data) with the event data in the third ring buffer **204-3**. In the upper part of FIG. 13, the second ring buffer **204-2** holds measured data D2 at times $t_{900}, \dots, t_{100}, t_{200}, t_{300}, \dots, t_{800}, \dots$ (100-millisecond interval) and the third ring buffer **204-3** holds event data D11 to D14 at times $t_{923}, t_{107}, t_{801}, \dots$ (aperiodic). However, relationships among the respective times are assumed to be $t_{900} < t_{923} < t_{100} < t_{107} < t_{200} < t_{300} < t_{800} < t_{801}$. In contrast to the above-described case in FIG. 12, the processor **302** of the data management device **300** corrects the date and time information of measured data using the date and time information of event data (that is, data with coarse temporal granularity) as a reference. The database **304** stores the measured data with the corrected date and time information in association with the event data used as a reference for correction.

(59) More specifically, in the example in FIG. 13, the date and time information of the measured data D2 at time t_{900} immediately before reference time t_{923} is corrected to the reference time t_{923} , associated with the event data D11 to D14 at time t_{923} , and stored in the database **304**. Similarly, the date and time information of the measured data D2 at time t_{100} immediately before reference time t_{107} and at time t_{800} immediately before reference time t_{801} is corrected to the reference times t_{107} and t_{801} respectively, associated with each event data D11 to D14 at time t_{107} and t_{801} , and stored in the database **304**. Note that the measured data D2 at times t_{200}, t_{300} or the like, which are not times immediately before the reference time is excluded from storage targets of the database **304** as a result of the time matching process. In this way, the database **304** can hold measured data and event data with uniform temporal granularity. This makes it easier to monitor and analyze measured data and event data.

(60) Note that in FIG. 12 and FIG. 13, while the measured data in the second ring buffer **204-2** and the event data in the third ring buffer **204-3** are temporally matched, it is also possible to temporally match the measured data in the second ring buffer **204-2** with the control bit data in the fourth ring buffer **204-4** using a similar process.

(61) FIG. 14 is a diagram illustrating another configuration example of the system **10** to implement the method according to the embodiment of the present invention. In FIG. 14, the system **10** is provided with a plurality of control devices **200**. A first control device **200A** is connected to some of the plurality of drive sections **120** and sensors **140** of the semiconductor manufacturing apparatus **100**, and a second control device **200B** is connected to some other parts of the plurality of drive sections **120** and sensors **140** of the semiconductor manufacturing apparatus **100**.

Functions of the respective components of the system **10** in FIG. 14 are the same as the functions of the respective components of the system **10** in FIG. 1.

(62) FIG. 15 is a diagram illustrating details of the process in step S205 of the flowchart in FIG. 2 applicable to the system **10** in FIG. 14. In the system **10** in FIG. 14, as described above, the first control device **200A** and the second control device **200B** respectively write high speed sampling data to the first ring buffer **204-1** periodically at the first temporal granularity (e.g., once every 10 milliseconds) and write low speed sampling data to the second ring buffer **204-2** periodically at the second temporal granularity (e.g., once every 100 milliseconds). Here, for example, when the time of the timer **207** of each control device **200** does not exactly match or when there is a difference in processing load among the respective control devices **200** or the like, date and time information actually stored in the first ring buffer **204-1** (or the second ring buffer **204-2**) of the respective control devices **200** can be misaligned.

(63) The data management device **300** performs the process of storing the data in the first ring buffer **204-1** (or the second ring buffer **204-2**) of each control device **200** as it is in the database **304** and temporally matching the data in the database **304** at an appropriate point in time (e.g., at the time of data monitoring or analysis) after that. Alternatively, prior to storing the data extracted from the first ring buffer **204-1** (or the second ring buffer **204-2**) of each control device **200** in the

database **304**, the data management device **300** performs the process of temporally matching the data.

(64) For example, the processor **302** of the data management device **300** corrects the date and time information of the second control device **200B** using the date and time information of the first control device **200A** as a reference. The measured data of the second control device **200B** with the corrected date and time information is stored in the database **304** in association with the measured data of the first control device **200A** used as a reference for correction. In the example shown in FIG. **15**, with the date and time information corrected to reference time **t100** of the first control device **200A**, the measured data **D10** at time **t098** of the second control device **200B** is associated with the measured data **D1** of the first control device **200A** at the reference time **t100** and stored in the database **304**. The measured data **D10** at times **t203** and **t301** of the second control device **200B** are likewise stored in the database **304** after correcting the date and time information to reference times **t200** and **t300** of the first control device **200A** respectively. Note that although the date and time information of the first control device **200A** is used as a reference here, the date and time information of the second control device **200B** may also be used as a reference. Which date and time information is used as a reference can be set in the data management device **300** in advance.

(65) The time matching process described with reference to FIGS. **11** to **13** may be applicable to the system **10** provided with the plurality of control devices **200** in FIG. **14**. In such a case, any one of the two ring buffers shown in FIGS. **11** to **13** may be the ring buffer of the first control device **200A** and the other may be the ring buffer of the second control device **200B**. For example, in the example in FIG. **11**, the date and time information of the low speed sampling data held in the second ring buffer **204-2** of the second control device **200B** can be corrected using the date and time information of the high speed sampling data held in the first ring buffer **204-1** of the first control device **200A** as a reference. Or, on the other hand, the date and time information of the low speed sampling data held in the second ring buffer **204-2** of the first control device **200A** can also be corrected using the date and time information of the high speed sampling data held in the first ring buffer **204-1** of the second control device **200B** as a reference. The same is applicable to the examples in FIG. **12** and FIG. **13** as well.

(66) Although the embodiments of the present invention have been described so far based on several examples, the aforementioned embodiments of the invention are intended to facilitate an understanding of the present invention, but not intended to limit the present invention. The present invention can be changed or improved without departing from the spirit and scope of the present invention, and it goes without saying that the present invention includes its equivalent components. Furthermore, within a scope in which at least some of the aforementioned problems can be solved or within a scope in which at least some effects can be exerted, any combination or omission of the respective components described in the claims and the specification is possible.

REFERENCE SIGNS LIST

(67) **10** system **100** semiconductor manufacturing apparatus **120** drive section **140** sensor **200** control device **202** processor **203** memory **204-1** first ring buffer **204-2** second ring buffer **204-3** third ring buffer **204-4** fourth ring buffer **205** input/output interface **206** communication interface **207** timer **300** data management device **302** processor **303** memory **304** database **400** communication wiring **500** network

Claims

1. A data management method for a semiconductor manufacturing apparatus comprising a plurality of sensors configured to measure operation of the semiconductor apparatus, the method comprising: acquiring, by a control device, data related to operation of the semiconductor manufacturing apparatus; identifying, by the control device, first data indicative of measured values collected from at least one of the plurality of sensors and second data of the acquired data; writing,

by the control device, the first data to a first ring buffer at first temporal granularity and the second data to a second ring buffer at second temporal granularity coarser than the first temporal granularity, wherein the second temporal granularity is based on the second data being written to the second ring buffer based on an aperiodic event time causing a change in the second data and wherein the first temporal granularity is based on a sensor measurement sampling time; and extracting, by a data management device, the first and second data from the first ring buffer and the second ring buffer respectively and storing the first and second data in a database.

2. The data management method according to claim 1, wherein the first data and the second data are written to the first ring buffer and the second ring buffer periodically at the first temporal granularity and the second temporal granularity respectively.

3. The data management method according to claim 1, wherein the step of storing the first and second data in the database comprises: correcting time information of the second data using time information of the first data as a reference; and storing the second data with the corrected time information in the database in association with the first data used as the reference for the time information.

4. The data management method according to claim 3, wherein one piece of the second data is associated with a plurality of pieces of the first data.

5. The data management method according to claim 1, wherein the step of storing the first and second data in the database comprises: correcting time information of the first data using time information of the second data as a reference; and storing the first data with the corrected time information in the database in association with the second data used as the reference for the time information.

6. The data management method according to claim 5, wherein a selected one of the plurality of pieces of the first data is associated with one piece of the second data.

7. The data management method according to claim 1, wherein the first data and the second data are measured data measured by a sensor provided in the semiconductor manufacturing apparatus.

8. The data management method according to claim 1, wherein the first data is measured data measured by a sensor provided in the semiconductor manufacturing apparatus, and the second data is data indicating operation contents of the semiconductor manufacturing apparatus.

9. A data management method for a semiconductor manufacturing apparatus comprising a plurality of sensors configured to measure operation of the semiconductor apparatus, the method comprising: acquiring, by a first control device, first data related to operation of the semiconductor manufacturing apparatus and indicative of measured values collected from at least one of the plurality of sensors; acquiring, by a second control device, second data related to operation of the semiconductor manufacturing apparatus; writing, by the first control device, the first data to a first ring buffer at first temporal granularity wherein the first temporal granularity is based on a sensor measurement sampling time; writing, by the second control device, the second data in a second ring buffer at second temporal granularity coarser than the first temporal granularity wherein the second temporal granularity is based on an aperiodic event time causing a change in the second data; extracting, by a data management device, the first and second data from the first ring buffer and the second ring buffer respectively; correcting, by the data management device, time information of the second data using time information of the first data as a reference; and storing, by the data management device, the second data with the corrected time information in a database in association with the first data used as a reference for the time information.

10. A control device connected to a semiconductor manufacturing apparatus, the control device comprising: a plurality of sensors configured to measure operation of the semiconductor apparatus a first ring buffer; and a second ring buffer, wherein the control device is configured to: acquire data related to operation of the semiconductor manufacturing apparatus; identify first data indicative of measured values collected from at least one of the plurality of sensors and second data in the acquired data; write the first data to a first ring buffer at first temporal granularity wherein

the first temporal granularity is based on a sensor measurement sampling time; and write the second data to a second ring buffer at second temporal granularity coarser than the first temporal granularity wherein the second temporal granularity is based on an aperiodic event time causing a change in the second data.
